Designer and Security Professional dedicated to being of service to others.

MY STORY

I am a freelance industrial designer and security professional, but more importantly I am a creative thinker and storyteller who values curiosity and empathy.

My experience in design and security give me a unique advantage in UX/UI design. Working in security trained me to be an observational researcher. Daily, I analyzed each tenant or guest that would interact with our rules, our sign in device at the front desk and to the building's architectural design.

Today I translate that deep understanding of human factors, safety and spatial navigation along with my design knowledge to create effective solutions to design problems.

DIGITAL DESIGN TOOLS

UX/UI
Figma | Wireframing
Information Architecture
Interaction Design
User Research
Adobe Creative Suite

Spatial and 3D Design
Rhino | Solidworks | Fusion
Onshape | CAD Drafting | Keyshot

Technical Foundations
HTML5 | JavaScript
CSS3 | Github

PHYSICAL DESIGN

Hand Artist
Sketching | Marker Rendering
Concept Ideation

Materials Knowledge
Wood | Metal | Glass | Clay
Plastics & Polymers | Foam
Vacuum Forming | CNC
Steam Bending | Mold Casting
Rapid Prototyping
3D Printing | Isometrics
Orthographics | Table Saw

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01

THESIS PROJECT

2020

HearRing Gem

Beingdeafishard. Manyof those inflictedwith deafness struggle withintheir own home torecognize important sounds thathelp to navigatelife.The HearRing Gem is a product for the deaf. Thisdevice alerts those with hearing loss of important sounds around their home by flashing an LED light, displaying icons and/or vibrating.

LIVING WITH DEAFNESS

Manydeafindividualscanstruggle to recognize important sounds within their own home.

SENSORY REALLOCATION

The brain in deaf individuals adapts to deafness by allocating resources to focusing on visual information.

VISUAL ADAPTION

Deaf brains adapt by focusing on visual attention, particularly with motion detection, peripheral vision and spacial attention.

How might we make identifying sounds within a home visually easier for the deaf community?

https://pmc.ncbi.nlm.nih.gov/articles/PMC2885708/

HOME SOUNDS

Let’s design around the average home lifestyle.

Kitchen Appliances

Doors

Washers and Dryers

Voices

Alerts

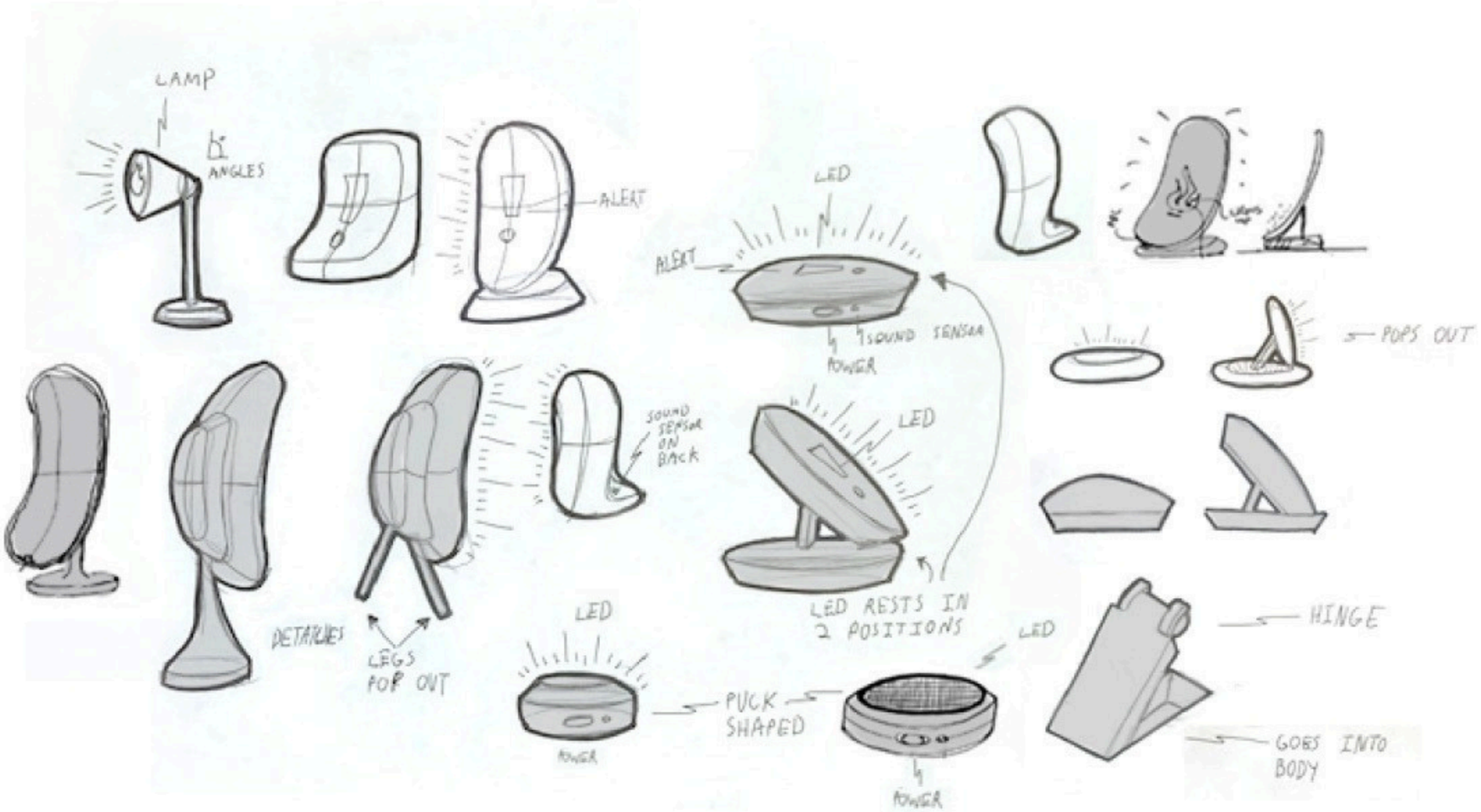
Pets

Let’s create a smart device that alerts the user of common household sounds through LED alerts and vibrations.

INSPIRATION

IDEATIONS

- Clamshell Design
- LED Screen
- Fits in Pocket
- Customize Alerts



Fits into the palm of your hand, pockets, under your pillow and can be mounted to your wall.

Backstand allows user to adjust the LED screen into upright position

Uses USB-C charger.

Housing is injected molding UL94-V0 rated ABS/PC Blend.

Uses MEMS microphones.

Microphone port has acoustic mesh to protect MEMS sensor from dust and moisture.

Record sounds for the HearRing Gem to recognize.

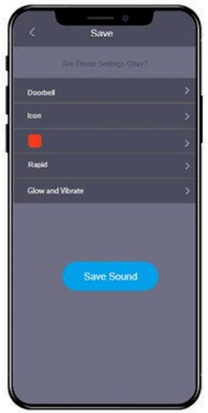
Customize alert icons, colors and speed for sounds.

Uses Digital Signal Processing to isolate important sounds from background sounds.

Uses ESP32-S3 microchip to connect to both WiFi and Bluetooth.

Uses WiFi to connect to other HearRing Gem devices in other rooms.

Uses bluetooth to connect to phone.

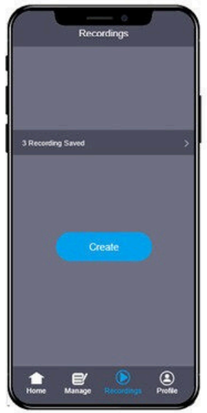


Save specific sounds and customize your alerts

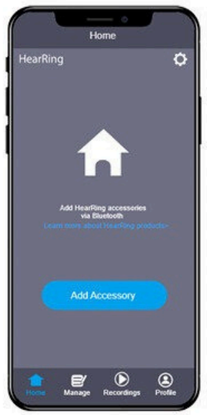
Choose what icon you want shown

Choose what color your device glows

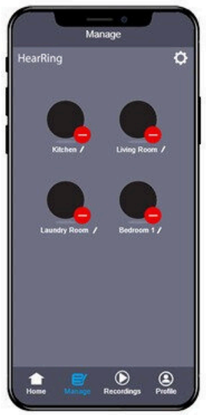
Choose how fast your device flashes



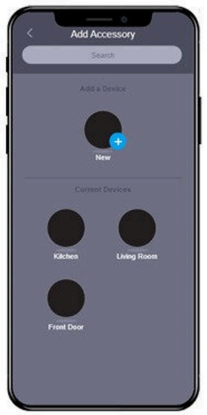
View your recordings



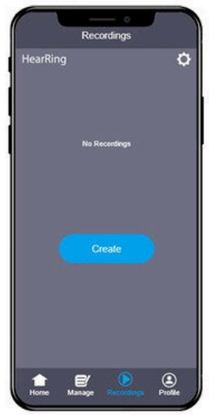
Sync HearRing Gems to the app



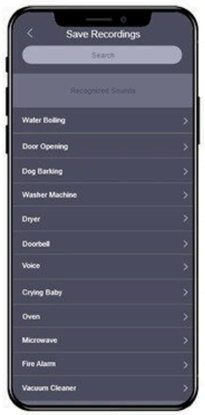
View exsisting devices



Manage or add a device



Save a sound for your device to recognize



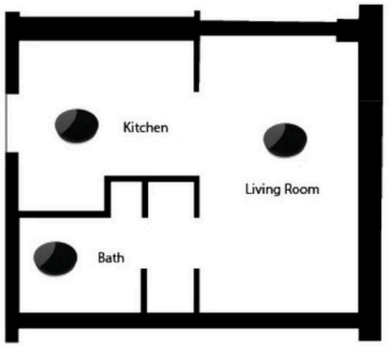
Choose from a number of recognized sounds

The user begins cooking food in the kitchen.

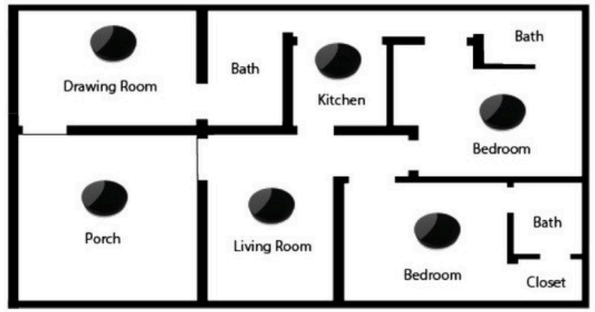
Someone's rings the user's doorbell.

HearRing Gem alerts user.

Apartment



House



HearRing

02

2024

AED Emergency Station

Emergency situations can be complicated and require quick decision making. The AED Emergency Station is a park station designed to help during such decisions. This station helps during heart attacks, fires and general emergencies where security or police involvement is necessary.

A MAJOR PROBLEM
There are **220,000 victims** of sudden cardiac arrest per year in the United States.

ACTING FAST MATTERS
Waiting for the arrival of emergency medical system personnel results in only **5-7% survival.**

THE BEST DEFENSE
Immediate defibrillation **increases survival rate up to 60%** one year after sudden cardiac arrest.

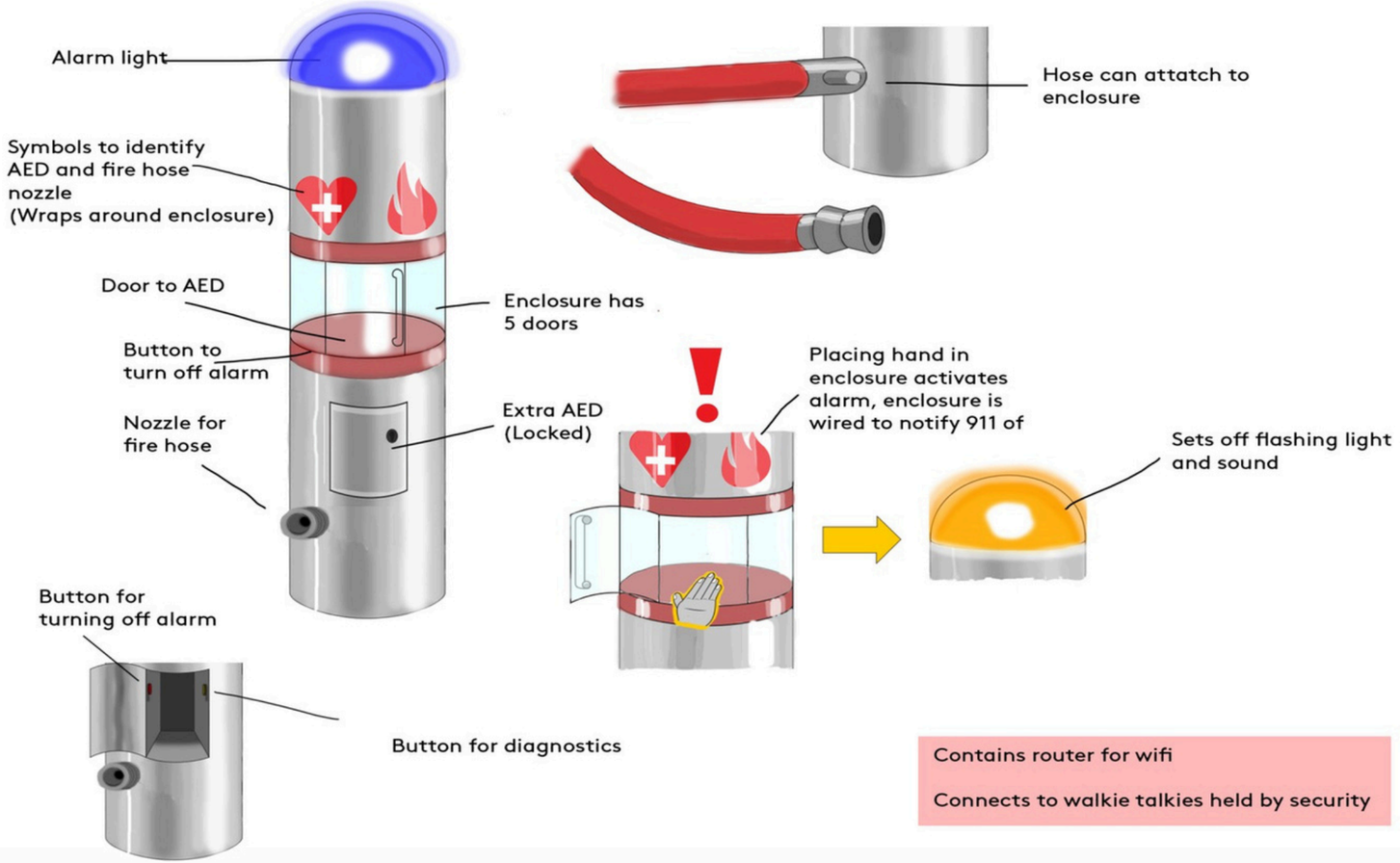
How might we design an emergency station that prioritizes Automated External Defibrillators?

Source: Osha Saving Sudden Cardiac Arrest Victims in the Workplace - https://www.osha.gov/sites/default/files/publications/osha3185.pdf

INSPIRATION

IDEATIONS

- Locked Enclosure
- Fire Hydrant
- Alarm Light
- Extra AED



Uses bright colors to make understanding the station easier during high stress emergencies.

A fire extinguisher is locked in a breakable case for fire emergencies.

Station made out of Fiberglass Reinforced Polyester.

The emergency station has a built in fire hydrant at the back of the station.

Positioning of AEDs, extinguisher and hydrant allow for easy access even if at the same time

Fire hydrant is made out of cast iron.

Uses bright colors to make understanding the AED easier during high stress emergencies.

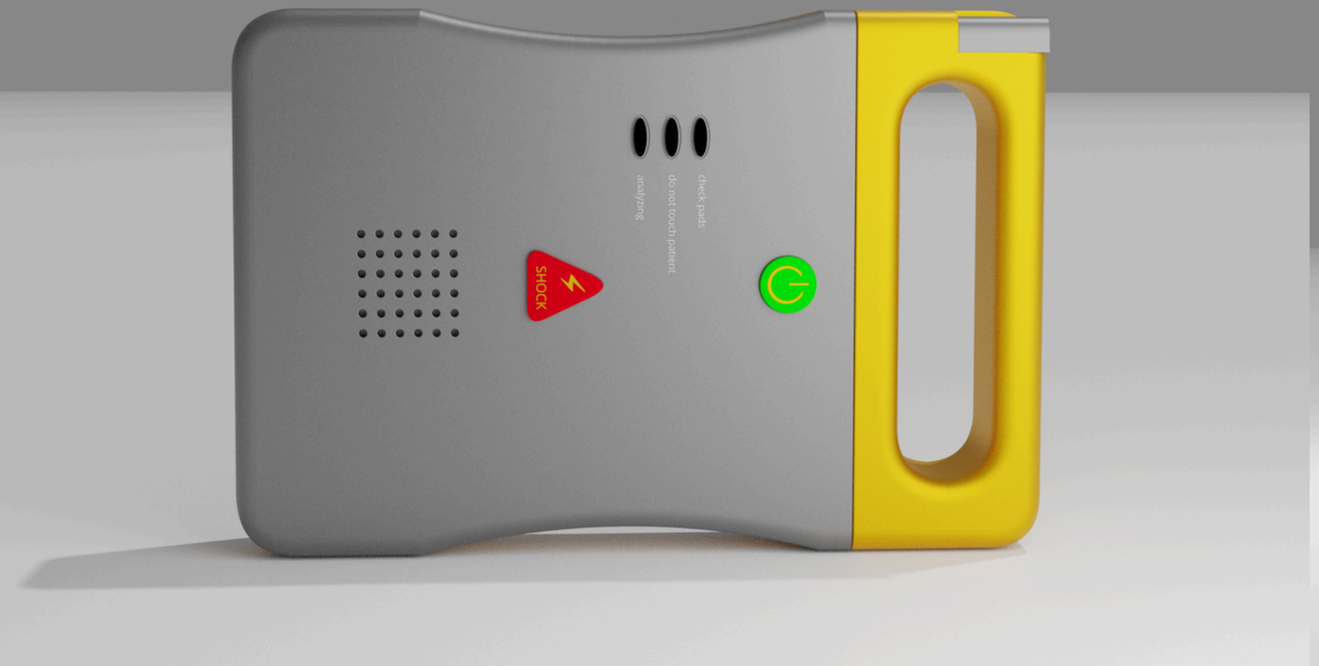
Easy to grab and pull out of station and locker.

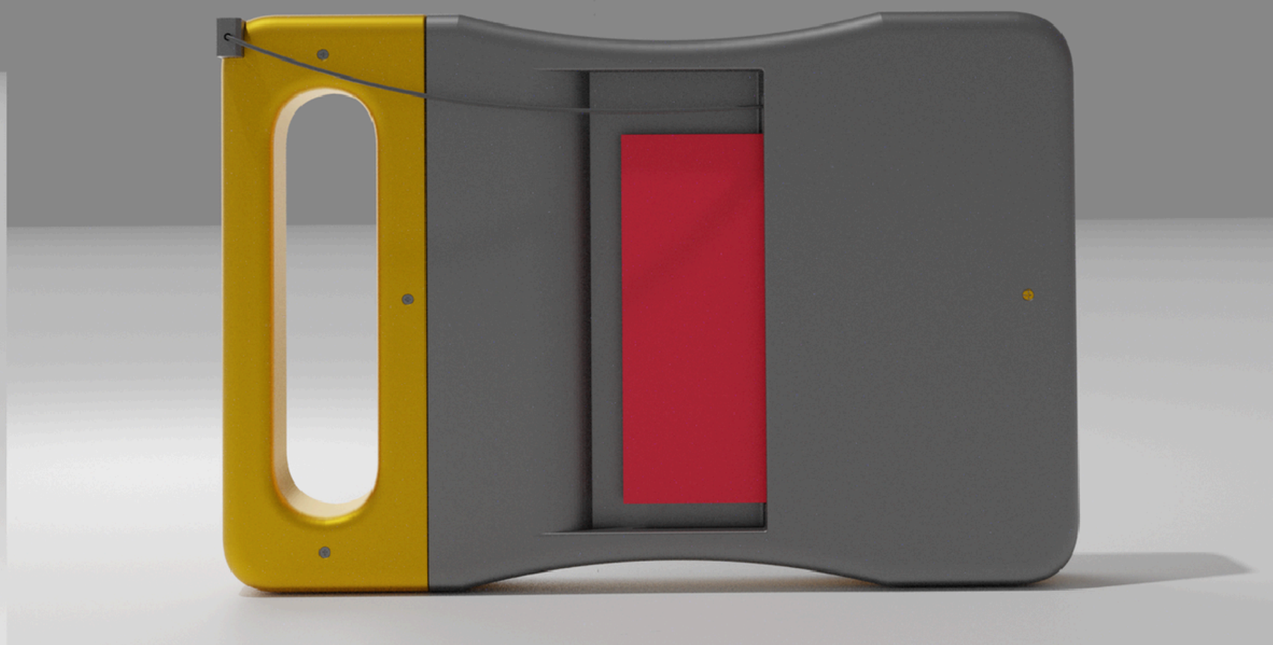
Front of AED gives instructions during use through speaker and LED.

Back of AED contains pads in a red packet.

Red packet includes instructions on how to operate the device.

AED housing is a high impact PC/ABS blend.

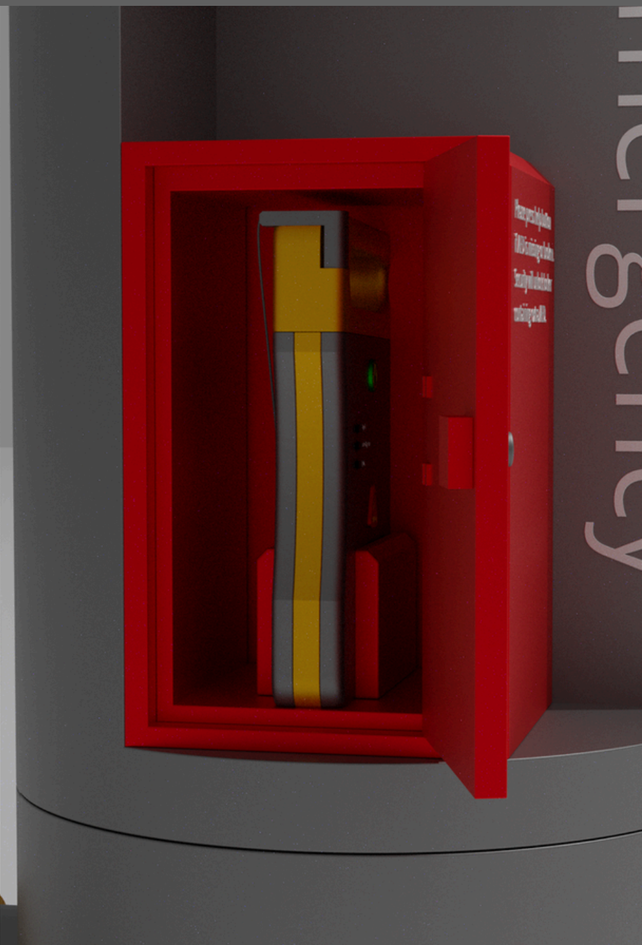




Call 911 to retrieve a code to unlock the top AED.

The Emergency Station contains an extra AED that can only be opened by security.

Yellow button will contact security and allow security to speak with the user via speaker.



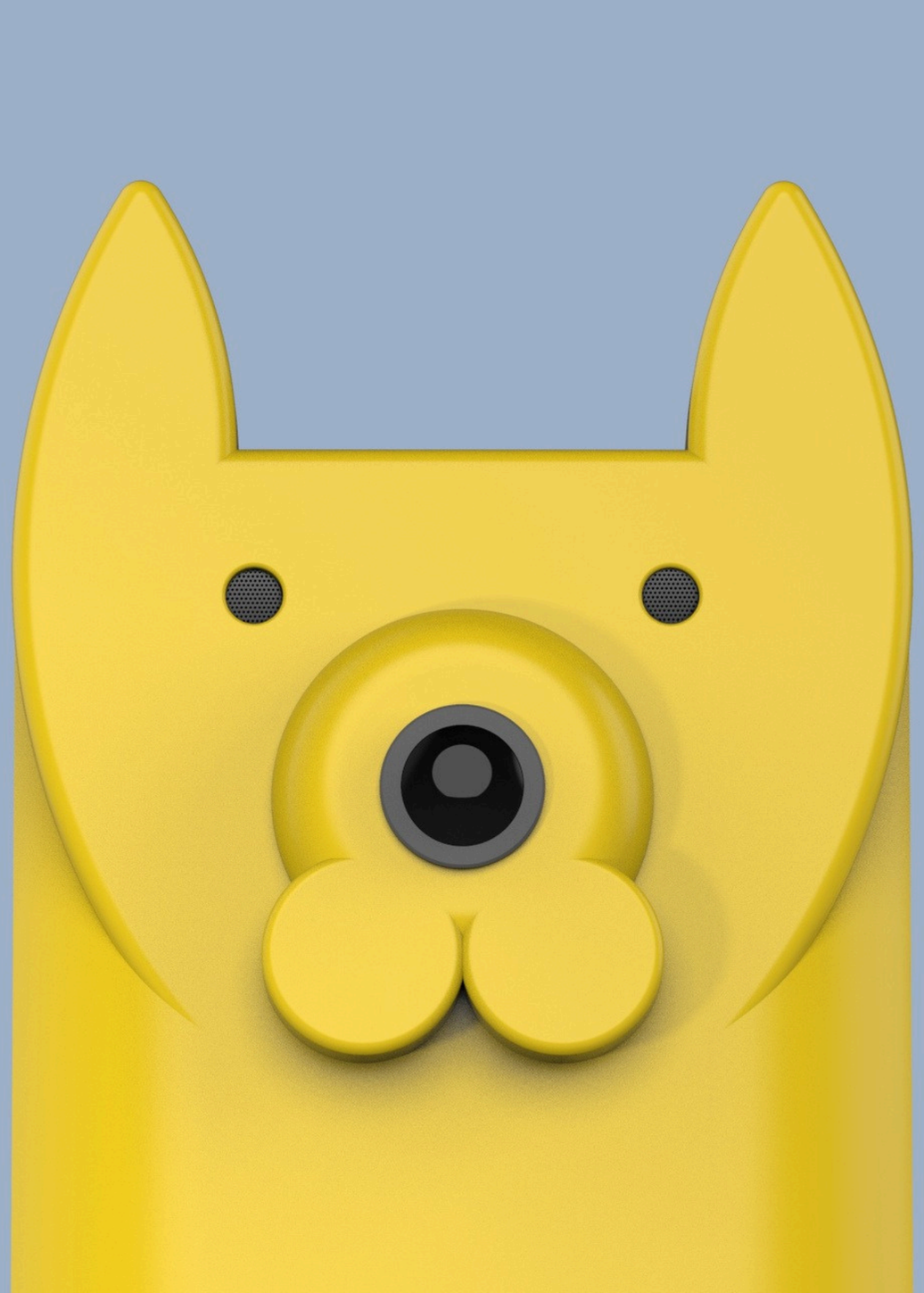
AED
01 Emergency
Please call 911
to unlock AED
Press for Assistance
Please press help button
if AED is missing or broken.
Security will unlock locker
containing extra AED.

03

2024

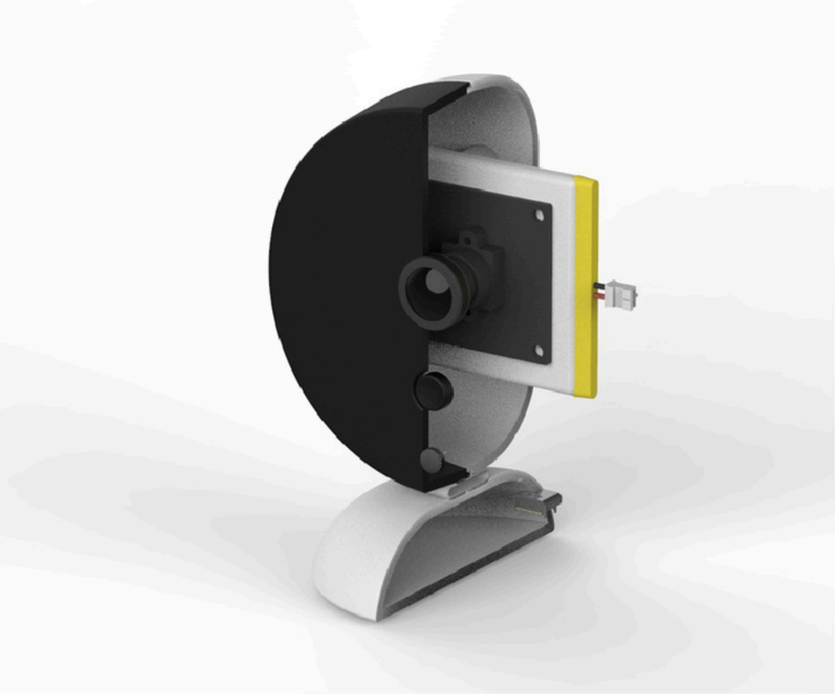
DogBox

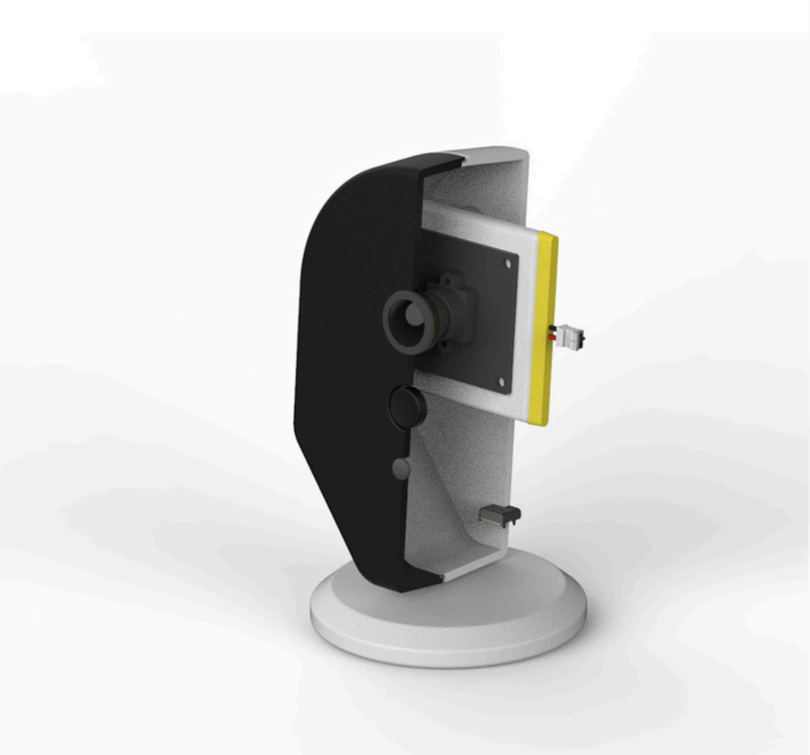
DogBox is a fun alternative to the pet feeder with a camera. There are a lot of similar products on the market, however DogBox is designed to be playful, look great in your home and protect your pet.



This project started with an exploration on camera components.

I explored different camera shell designs to see how the components fit.





RESPONDING TO COLOR
New study showsthatdogs prefer and respond more to the color yellow over any other color.

THE NEED FOR ROUTINE
Being fed consistently helps support your dogs metabolism, digestion and mental health.

ALWAYS LISTENING
Dogs have been found to respond to human words, especially when referred to by name and while using “baby talk”,

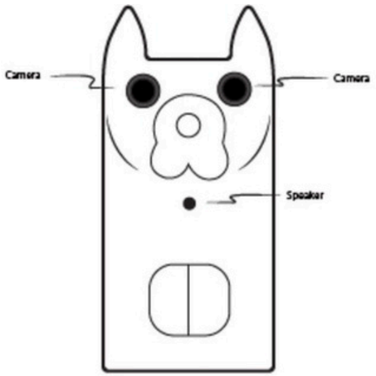
How might we redesign a dog feeder that will keep a dog’s schedule even when their owner is away through the use of auditory and visual cues?

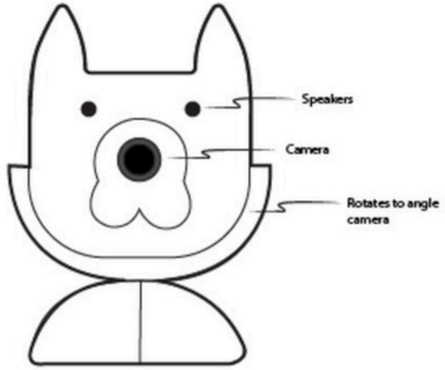
https://phys.org/news/2025-04-ears-reveals-dogs-remarkable-ability.html
https://bondvet.com/blog/daily-dog-routine
https://pmc.ncbi.nlm.nih.gov/articles/PMC2885708/

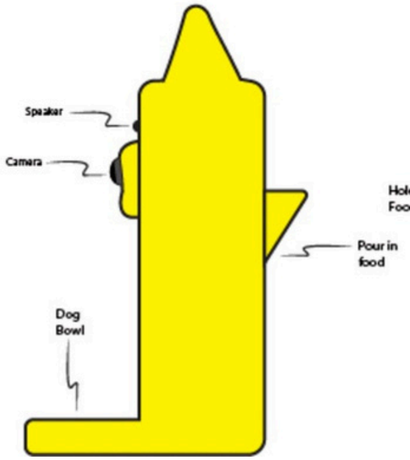
INSPIRATION

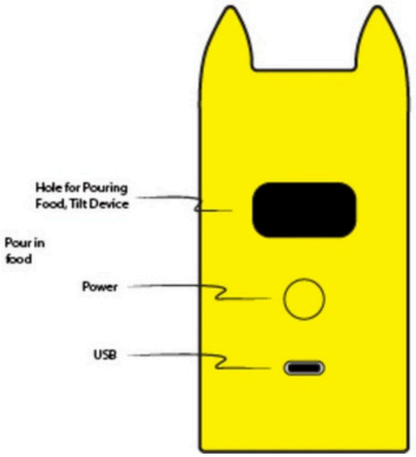
IDEATIONS

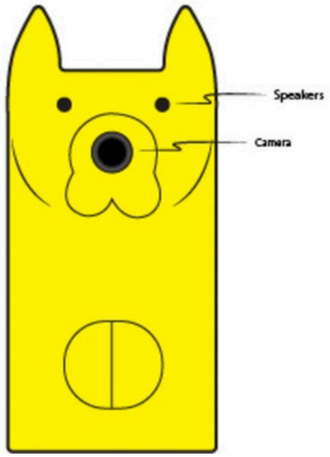
- Colorful
- Camera Nose and Speaker Eyes
- Sliding Door Mechanism
- Pour Food in Back to Fill











DogBox has a sliding door mechanism dispense food.

Contains one camera for its nose to monitor your pet.

Contains two speakers for eyes allowing you to speak to your pet.

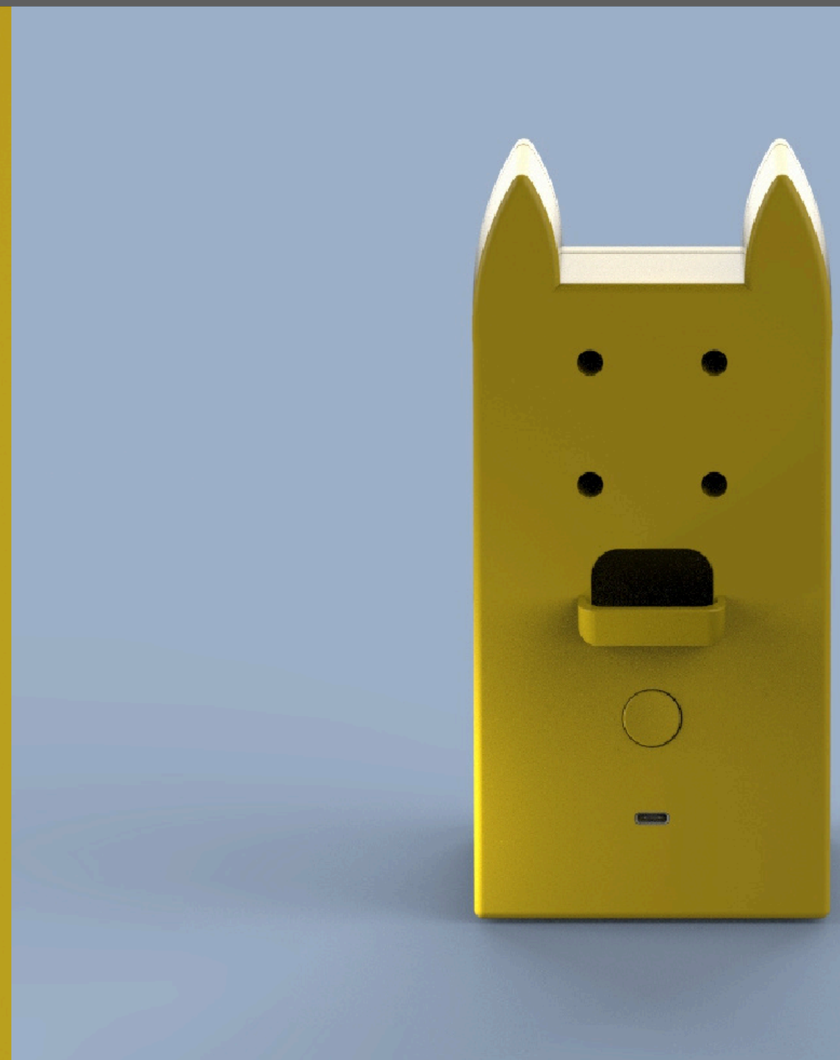
Front shell is stainless steel for a more hygienic and safe experience for pet.

Back shell is ABS plastic.

DogBox. Charge the device with a USB-C charger.

The power button can be found at the back of the device.

Pour your dog’s food into the back to fill up your DogBox.



Connect to other DogBoxes in different rooms to track your pet’s activity.

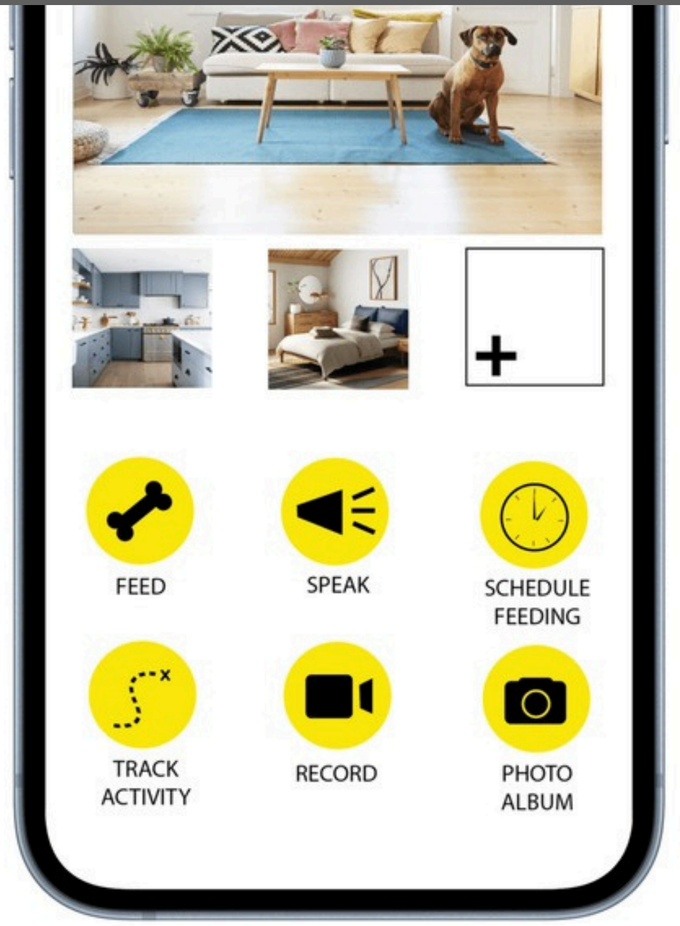
Schedule a feeding time.

Speak to your dog through the built in speakers

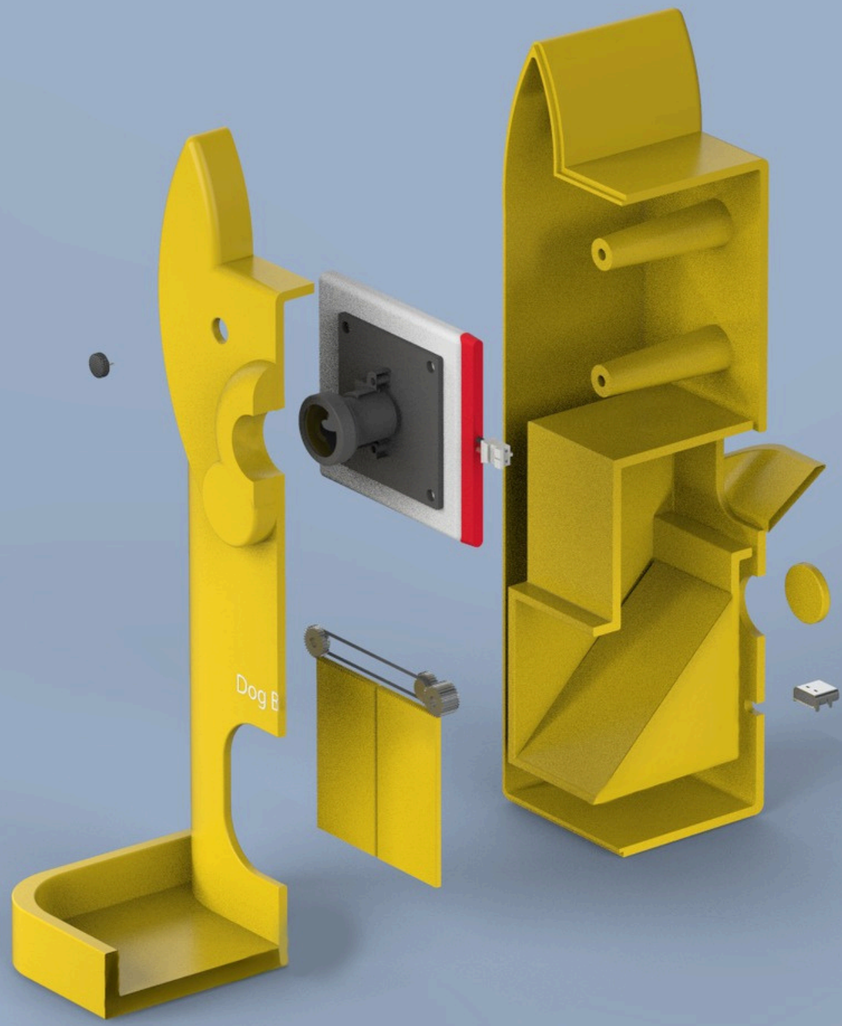
Activate the feeder

Speak to your pet anytime through the built in microphone

View history of which rooms your pet has traveled to



Schedule an automated feeding time


Dog B

Dog Box